Hello friends,

I hope you're all doing well.
The following exercises are not exam questions, but rather open-ended prompts whose answers can often be found online, through chatbots, or even in technical interviews. These are intended to encourage deeper exploration and self-directed learning based on your current knowledge.

We've selected these questions based on real-world experience and past exam patterns to help you better understand the types of concepts that appear in the LPIC exam. Many of the questions don't have a single correct answer — instead, they are designed to broaden your perspective and understanding.

Please be kind and patient with yourself as you write and research your responses. Please complete the following exercises using the same format as the directory structure below. Combine your answers into a single PDF file named answer.pdf **and submit a pull request.**

📁 Reference format:
https://github.com/devopsdoctors/Academy/tree/main/L1-JavanPahlevanan/Exercises/T(x)/name-family(emailAddress)/answer.pdf

📄 Sample answer file:
https://github.com/devopsdoctors/Academy/tree/main/L1-JavanPahlevanan/Exercises/T1/ali-farhadian(alifrd49@gmail.com)/answer.pdf

1. **Explain DMA (Direct Memory Access). How does it differ from NUMA (Non-Uniform Memory Access)?**
2. **Why is NVMe storage faster than traditional SSDs (SATA-based)?**
3. **What is the difference between SAS and iSCSI in terms of usage and architecture?**
4. **You have three disks (1 TB, 512 GB, and 256 GB). Design a custom server partition layout with separate mount points for `/`, `/boot`, `/var`, and `/home`.**
   How does this design affect scalability and future disk expansion?
5. **Provide the support matrix and key characteristics of the following file systems:**
   - NTFS
   - exFAT
   - FAT32
   - ext4
6. **What is the Linux Device Mapper, and what is it used for?**
7. **Explain the differences between RAID 0, RAID 1, RAID 5, RAID 6, and RAID 10. Include diagrams.**
8. **Explain the OSI 7-layer model and compare it with the TCP/IP 4-layer model. Include diagrams.**
9. **Assume traffic flows from Node A to Node D, with Node B and Node C in between:**
   - Which nodes use **MAC addresses** for communication?
     - Which OSI and TCP/IP layers handle this addressing?
   - Which nodes use **IP addresses** for communication?
     - Which OSI and TCP/IP layers handle this addressing?
   - Which nodes use **port numbers** for communication?
     - Which OSI and TCP/IP layers handle this addressing?

10. **Identify the OSI and TCP/IP layers for the following protocols:**

    - DNS
    - TCP
    - TFTP
    - UDP
    - HTTP
    - TLS
    - ARP
    - DHCP
    - ICMP
    - SNMP
    - CDP

11. **Define the following data units and specify which OSI/TCP-IP layers they belong to:**
    - Packet
    - Segment
    - Frame
    - Bit

12. **What are MTU and MSS? How do they differ?**
13. **How is MTU size calculated?**
14. **Why must MTU size be fixed? (Hint: fragmentation issues)**
15. **Explain how the ARP protocol works. Provide a step-by-step summary.**
16. **What is the difference between:**
    - Well-known ports (system ports)
    - Registered ports (user ports)
    - Dynamic / ephemeral ports
17. **Run `mtr` or `traceroute` to `1.1.1.1` and explain each hop in the output.**